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BE-CEM-MRO

CRANEBOT v2.0 ROBOTIC SYSTEM

Functional Specifications

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Page 2 of 16

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TABLE OF CONTENTS

1.	Introduction	4
2.	System description	5
2.1	Robot overview	5
2.2	Crane hook adapters	7
2.3	Rotational joint.....	8
2.4	On-board camera system	9
2.5	Transport trolley	11
2.6	Robot dimensions	12
3.	Manipulators	14
3.1	Grippers and tools	15

1. Introduction

This manual is intended to provide the functional specifications of the CRANEbot v2.0 teleoperated robotic platform, managed by the BE-CEM group. The manual describes the functionality and the mechanical specifications of the robot, in terms of weight, working load limit (WLL), payload and arms reachability.

This teleoperated robot, raised by overhead cranes, allows performing work at height, carrying out procedures remotely with the required degree of care and precision, in radioactive areas not reachable by common terrestrial robots. It can thus replace human beings in especially dangerous situations and take over their work.

CRANEbot can be used for remote transport and manipulation of dangerous equipment, also for the inspection and investigation of unknown objects and locations. The robot supports the operator in the following tasks regarding dangerous items:

- Identification and processing of objects
- Use of various devices to dispose of dangerous items
- Reconnaissance, monitoring and investigation by means of the built-in cameras or special cameras
- Use of various electrical and mechanical tools
- Transporting, positioning and remotely controlled use of measuring equipment.



2. System description

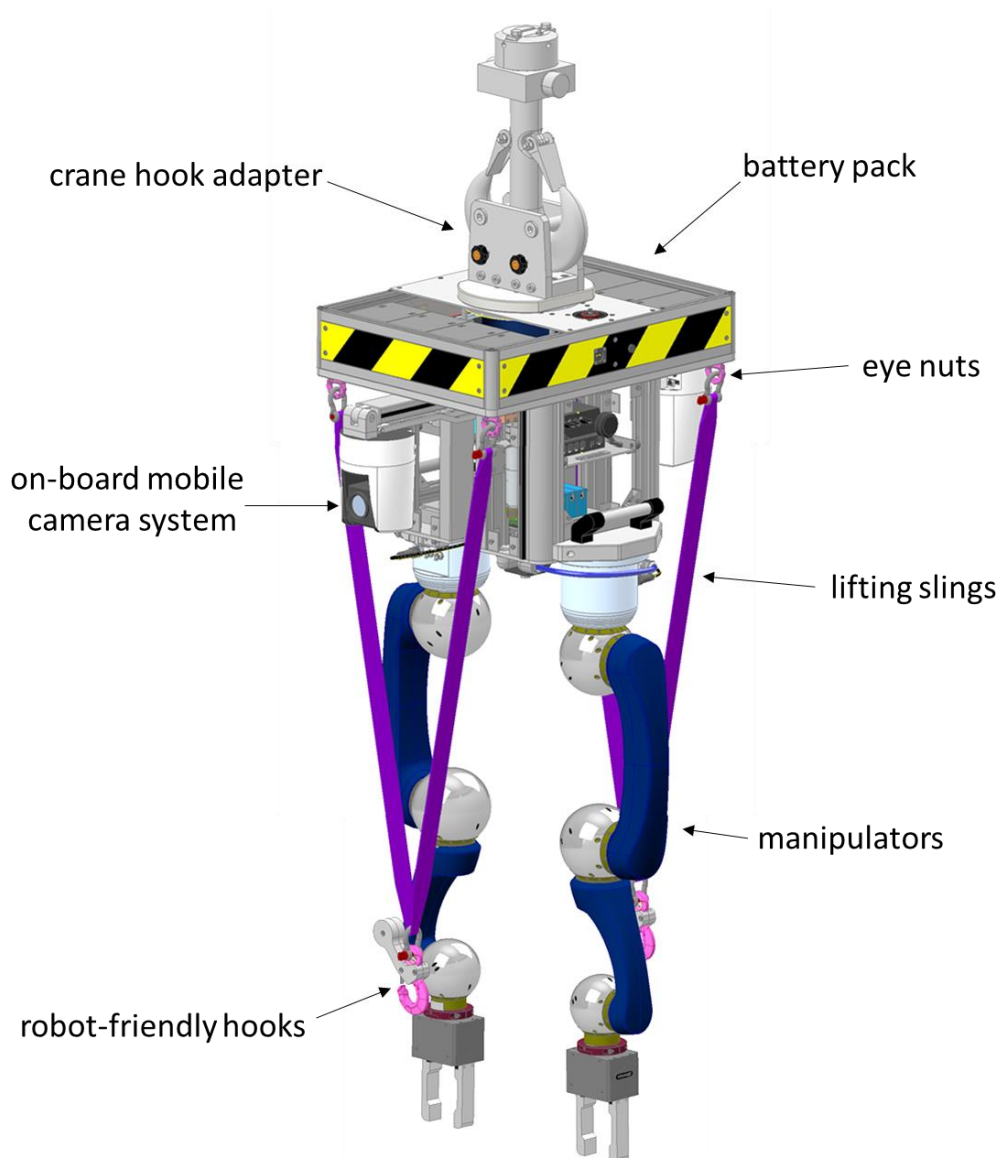
2.1 Robot overview

The central part of the robot consists of an aluminum frame that contains all the electronic devices and batteries needed to control the manipulators and the on-board camera system.

In the upper part, an adapter for double hooks DIN 15402 allows the lifting of the robot with overhead cranes.

Four RUD VRM M8 eye nuts can be used to hook and transport loads up to 500 kg.

Using two robot-friendly hooks, grabbed by the manipulators, it is possible to hook, transport, and release loads remotely.





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Page 6 of 16

CRANEbot v2.0 – Technical Data

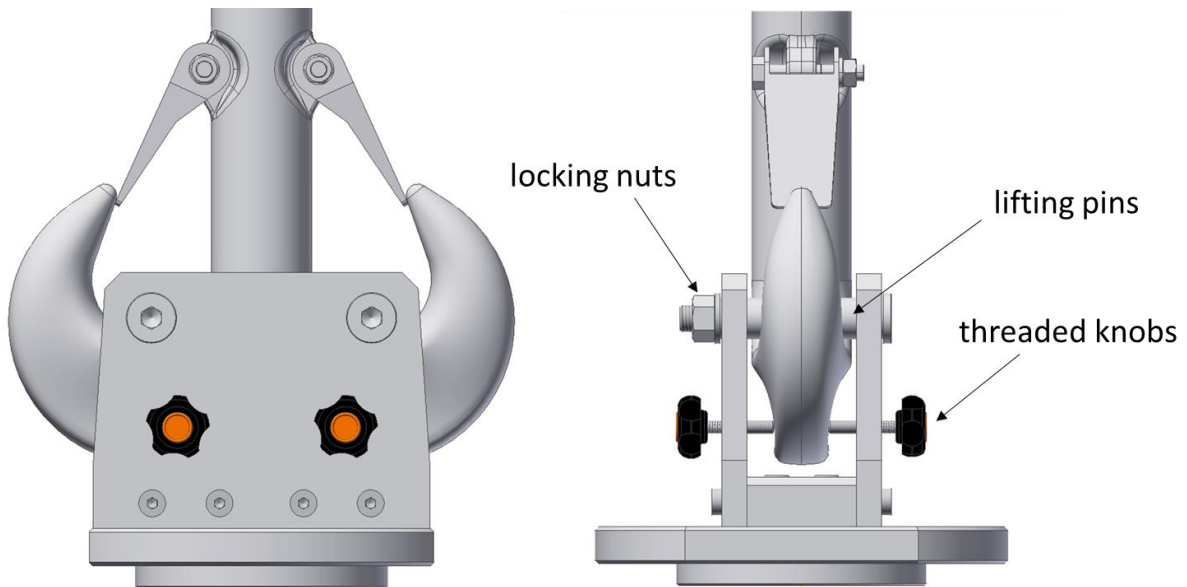
Weight [kg]	150
Working Load Limit (WLL) [kg]	500
Compatible crane hooks	DIN 15402 size 4,6,8,10
Platform rotation speed [rpm]	5
Manipulators	2x 6-axis SCHUNK Powerball
Power supply	8x Lead-gel rechargeable batteries 12V, 13Ah
Autonomy [h]	8 mixed operations, approx. 20 only camera operation
Cameras	2x PTZ zoom 30x 1x Fisheye
Network	Wi-fi, 4G

2.2 Crane hook adapters

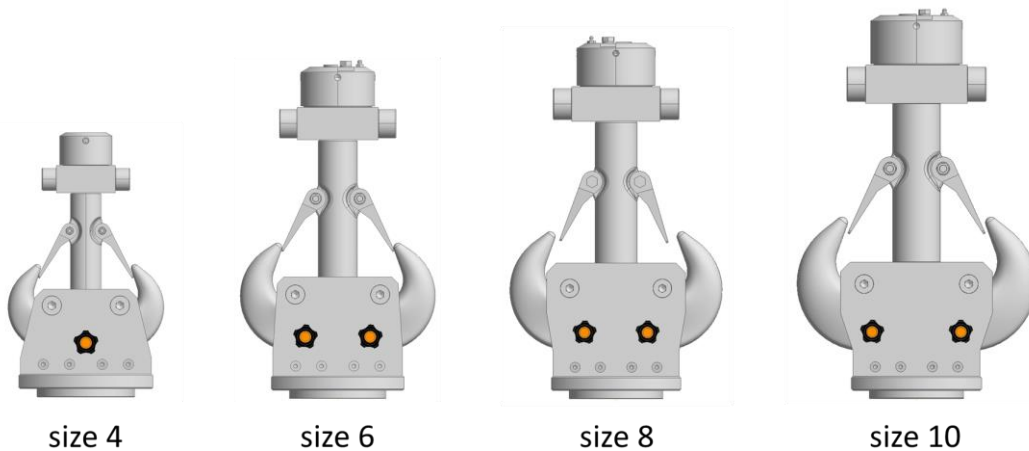
The crane hook adapters are lifting points designed to lift and move the platform using overhead cranes with DIN15402 (UNI 9470/1) double hooks.

Each adapter is composed of three plates arranged in a U shape fixed on the top of the platform.

Once the crane hook is inserted between the two vertical plates, it is constrained by two $\varnothing 25\text{mm}$ lifting pins with M16 locking nuts. Four M8 threaded knobs allow to center the hook and lock its lateral sliding.



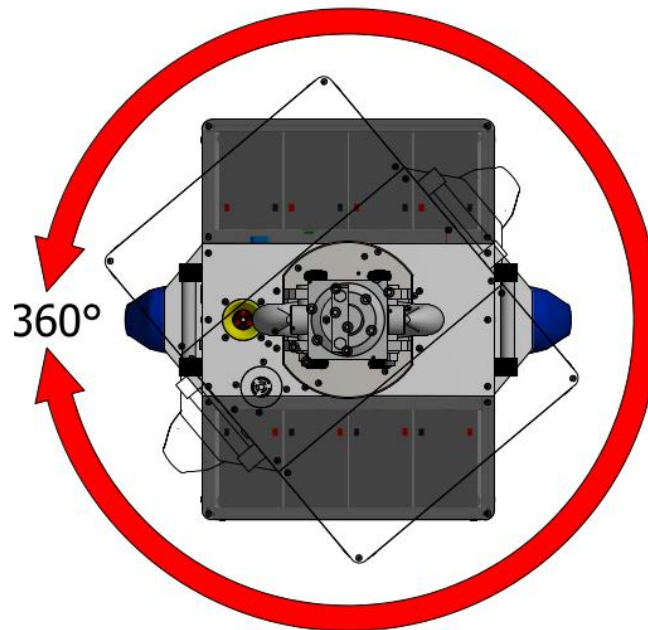
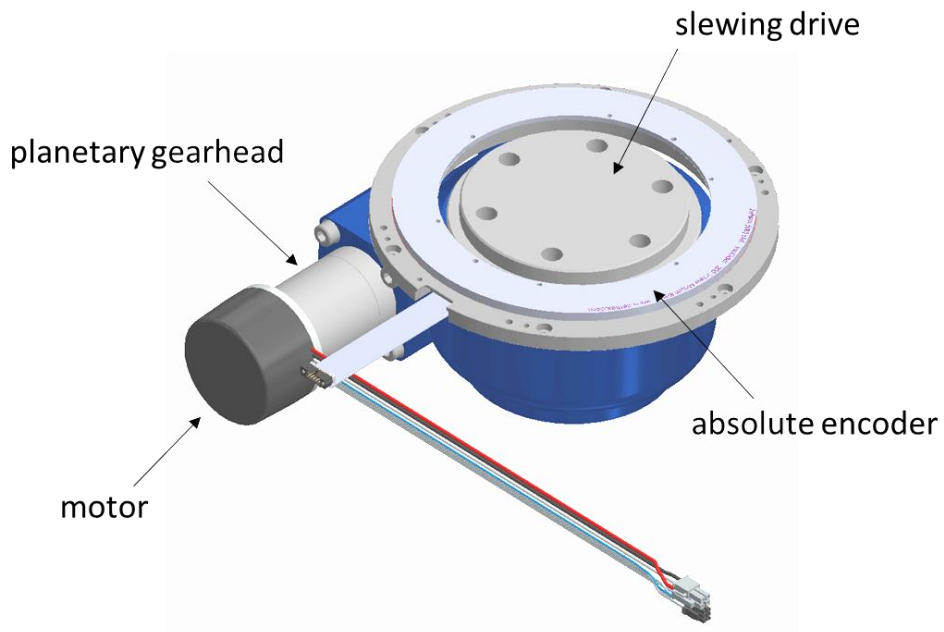
By replacing these parts, it is possible to adapt the lifting point dimensions to the hook size used in the intervention area. Four adapters are available for four different DIN15402 hook sizes: 4, 6, 8, 10.



For larger hooks it is necessary to harness the platform by tying two slings to the lifting pins.

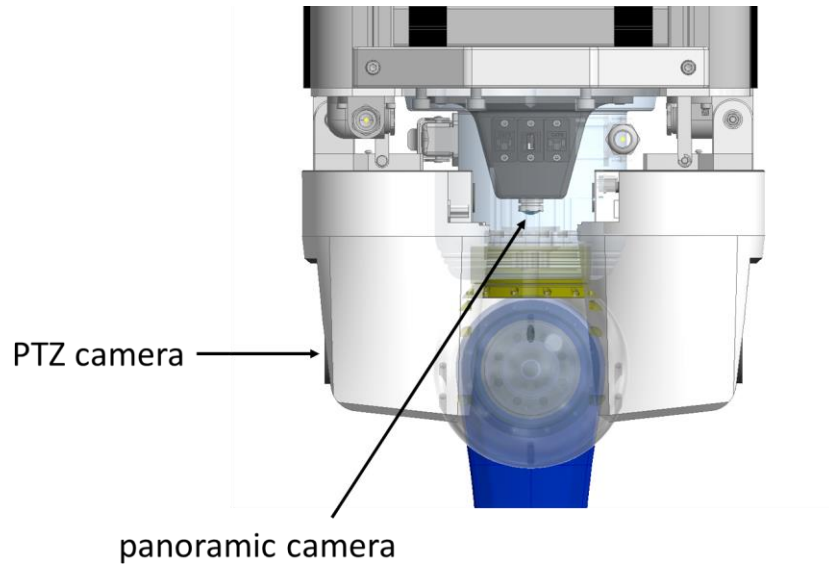
2.3 Rotational joint

The crane hook adapter is fixed on a rotational joint that allows rotating the platform and the load carried by it around the vertical axis of 360 degrees with a speed of 4 rpm. In order to use this feature, the crane hook rotation must be locked.

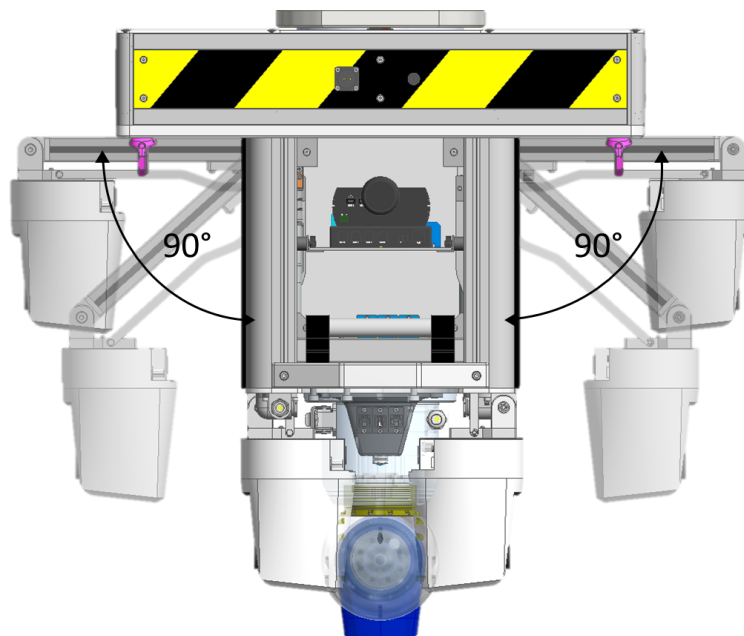


2.4 On-board camera system

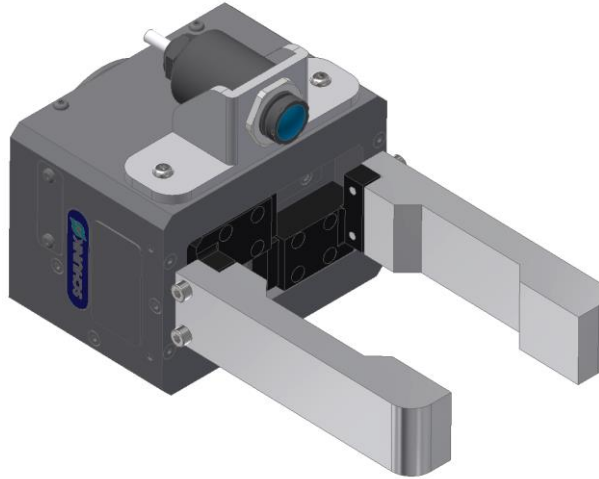
The CRANEbot on-board camera system consists of a panoramic camera (AXIS F1035-E Sensor Unit), located in the lower central part of the frame, and two PTZ cameras (AXIS V5925) located on the sides.



A seagull wings system allows the operator to independently move the PTZ cameras on the sides of the platform, improving the point of view during handling operations:



It is possible to install up to three additional cameras (AXIS F1005-E Sensor Unit) to have further points of view on the manipulator end-effectors.



AXIS V5925 PTZ Camera - Technical Data



Pan	$\pm 170^\circ$
Tilt	-20° - 90°
Zoom	30x optical, 12x digital
Resolution	1920x1080 HDTV
Weight [kg]	1.5

AXIS F1035-E Sensor Unit - Technical Data



Horizontal angle of view	194°
Vertical angle of view	113°
Zoom	n/a
Resolution	1920x1200/1080
Weight [kg]	0.096

AXIS F1005-E Sensor Unit - Technical Data



Horizontal angle of view	113°
Vertical angle of view	62°
Zoom	n/a
Resolution	1920x1080 HDTV
Weight [kg]	0.116

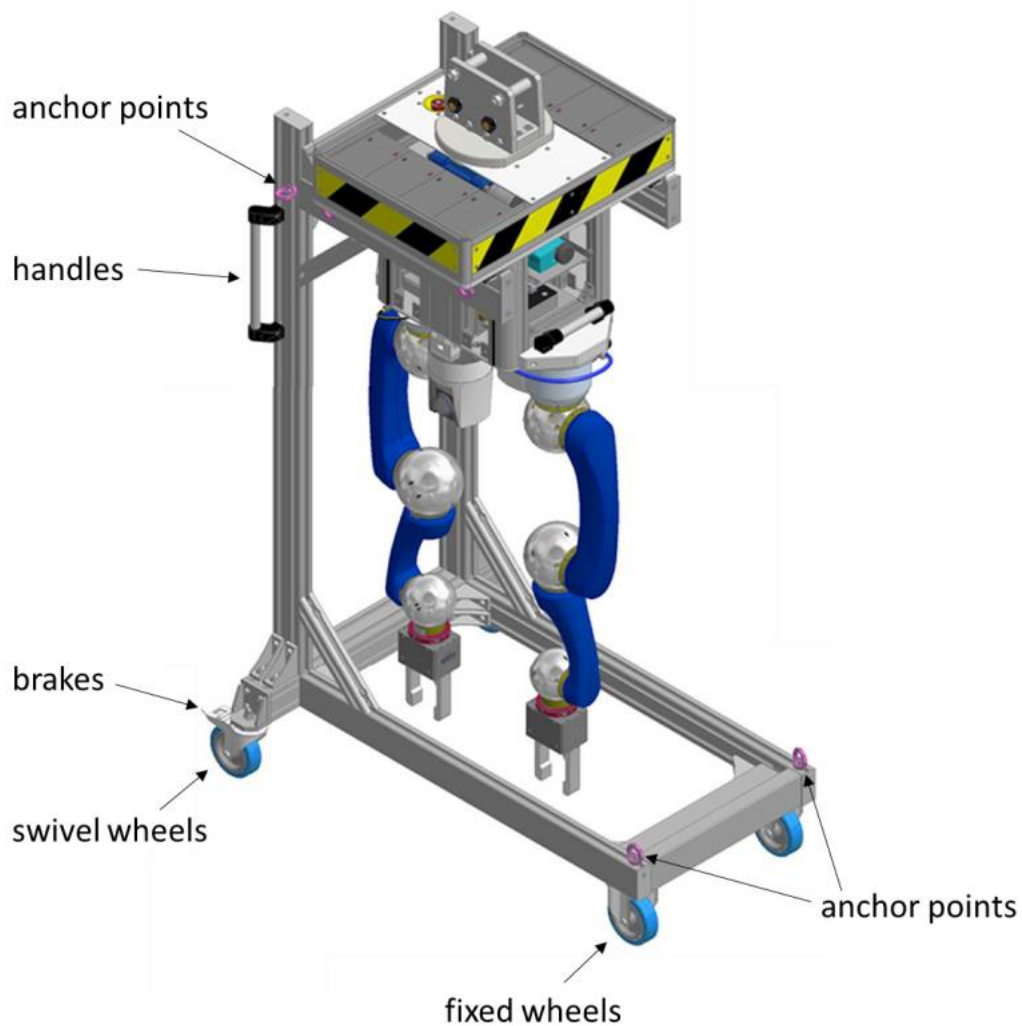
2.5 Transport trolley

Cranebot v2.0 is equipped with a trolley for transport to the intervention areas.

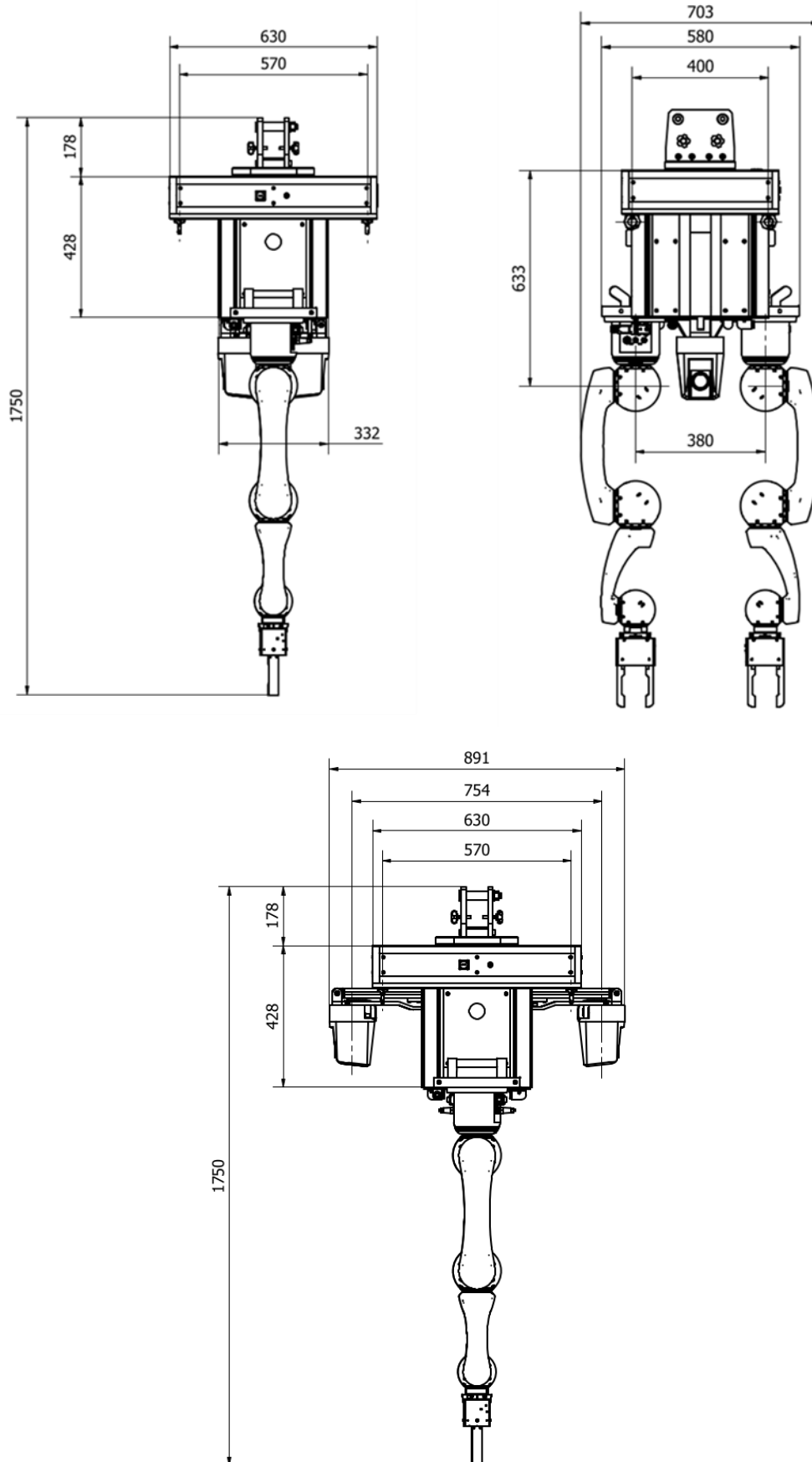
The trolley has a frame made of aluminum profiles, with two fixed front wheels and two swivelling rear wheels with brakes.

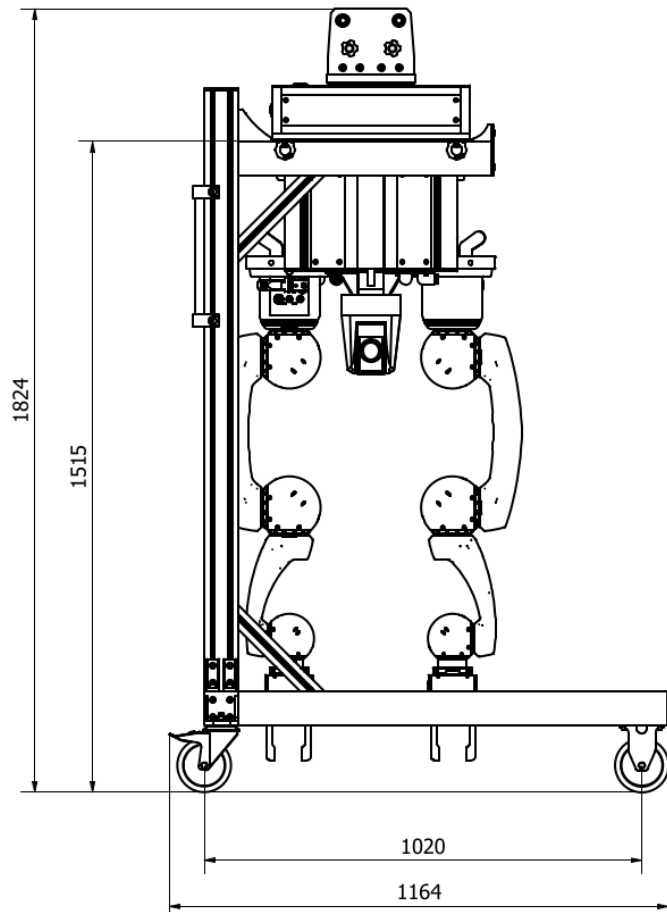
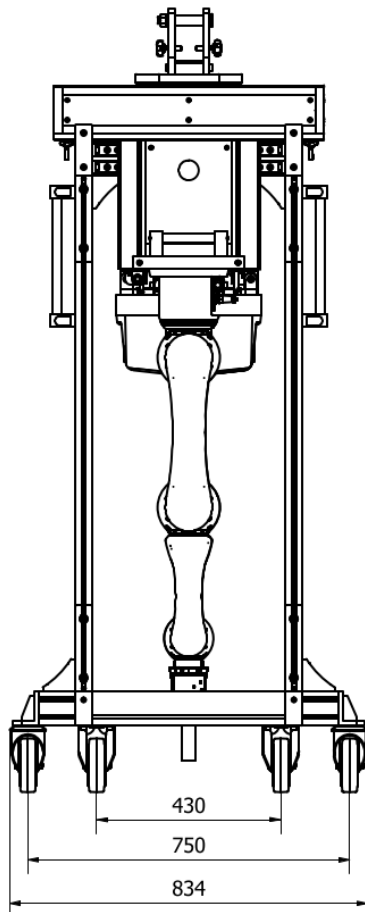
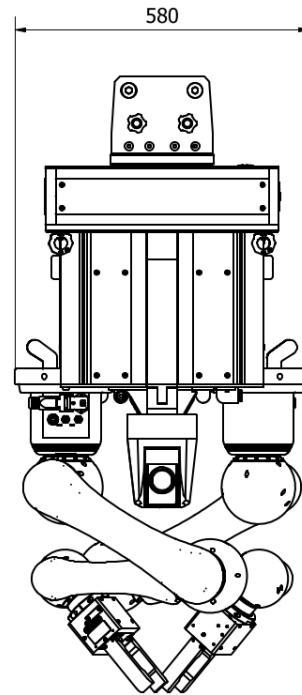
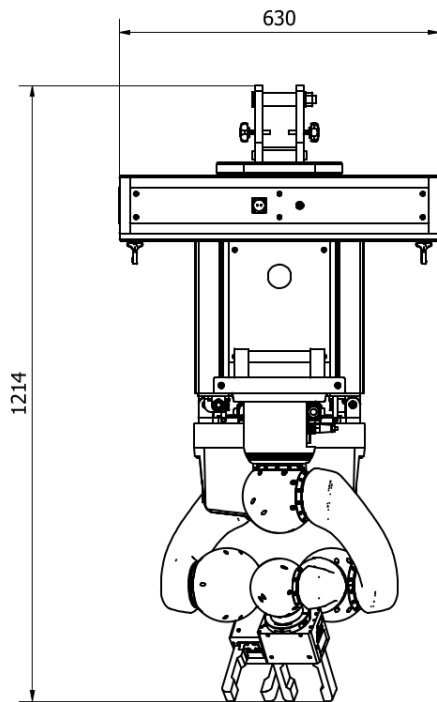
Four anchor points allow it to be harnessed during transport by truck.

Two handles can be used to push/pull the trolley by hand.



2.6 Robot dimensions

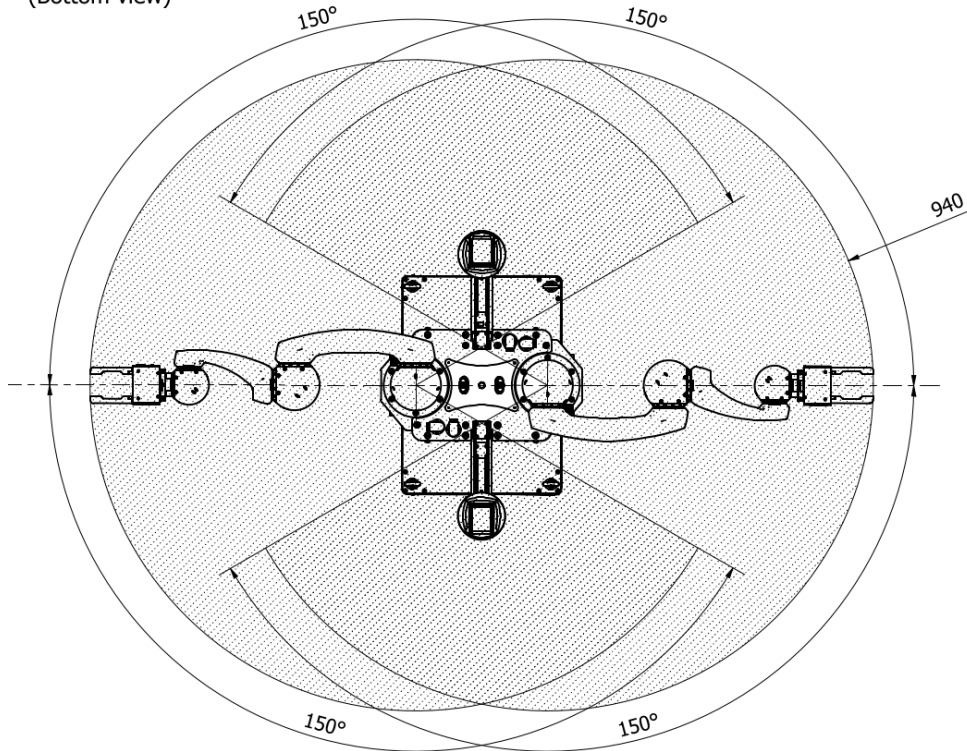




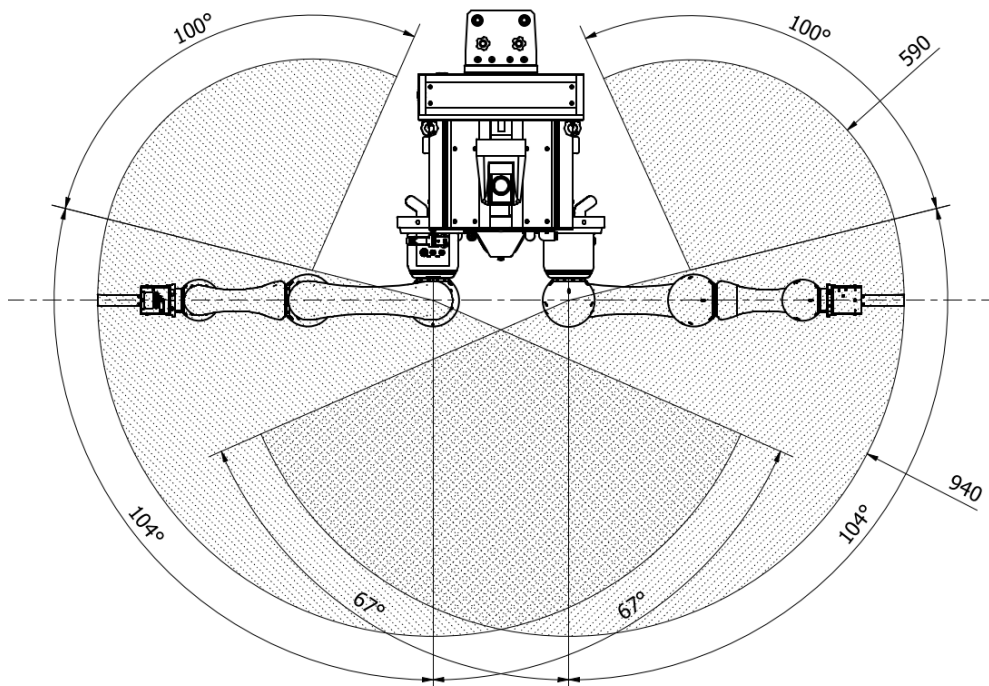
3. Manipulators

CRANEbot hosts two SCHUNK Powerball LWA 4P robotic arms

Working Space
(Bottom view)



Working space
(Side view)



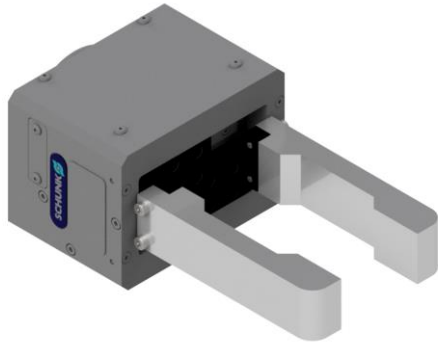
SCHUNK Powerball LWA 4P - Technical Data

	Type	6 DOF lightweight robot
	Number of axes	6
	Max. useful load [kg]	6
	Axes speed with nominal load [°/s]	72
	Repeat accuracy [mm]	±0.15
	IP class [IP]	40
	Weight [kg]	12.5

3.1 Grippers and tools

Depending on the operation to be carried out, each arm can be equipped with different accessories, including grippers, screwdrivers, RP sensors, cameras, endoscopes.

SCHUNK Finger Parallel Gripper PG-Plus 70 – Technical Data

	Stroke per finger [mm]	34
	Min./max. gripping force [N]	30/200
	Nominal force [N]	160
	Weight [kg]	1.2
	IP class [IP]	20
	Min./max. temperature [°C]	5/55

Impact Screwdriver – Technical Data

	Max torque [Nm]	205
	No load speed [rpm]	0-1000
	Tool holder [mm]	6.35
	Weight (with Li-Ion battery pack) [kg]	1.68
	Remote speed control	yes
	Remote torque control	n/a

Screwdriver – Technical Data



Max torque [Nm]	27
No load speed [rpm]	0-550
Tool holder [mm]	1.5-13
Weight (with Li-Ion battery pack) [kg]	1.8
Remote speed control	yes
Remote torque control	yes

ATOMTEX BDGK-24 RP Sensor – Technical Data



Energy Range	25 keV-10 MeV
Measurement range of ambient dose equivalent rate	30 nSv/h – 1 Sv/h
Limit of intrinsic relative measurement error	±20%
Response time for dose rate change from 0.1 to 1 µSv/h	≤2s
Overall dimensions [mm]	Ø60x200
Weight [kg]	0.5